
Computational Reproduction of a Multi-Path Photonic Experiment

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Repository: <https://github.com/jonathanssearch/RATISS-PHOTON> · label: computational only (no QPU)

Abstract

We reproduce, entirely in silico, the multi-path photonic experiment of Wen et al. (Science Advances 12, eaeh1011, 26/08/2026) using a paraxial engine implementing the reference equation itself. On the E-CANTON configuration, 8,396,800 equal-modulus paths yield a fidelity of 95.92% (two planes), 95.96% (three planes) and 96.77% (naive action), with intensity correlation 97.08–97.10%, a cross-check at 96.06% and a screen phase of 5.4° . These values fall inside the Canton window (95–98.5%). Crucially, at $T = 0$ K a single impact position out of 400 is observed: randomness *emerges* from the thermal bath, with no Born rule hard-wired.

Scientific question

Does the action of a paraxial world suffice to recover the measured multi-path statistics, without any hard-wired Born sampling?

Reference work

Wen et al., Science Advances 12, eaeh1011 (26/08/2026): 1,419,857 paths, fidelity $87.6 \rightarrow 98.5\%$, MAPE $8.17 \pm 3.50\%$. This work is not in competition with the reference: they study the real device, we study the world-model.

Method

Paraxial engine v2 implementing the paper's equation. The world action is quadratic: $k_0 dz + k_0 dy^2/(2 dz)$. Seed 20260927; full campaign runs in about one second. All visualisations use embedded Plotly and kaleido (laboratory decision C2: never hand-rolled 3D). The study is labelled computational-only; no quantum hardware was used.

Results

- E-CANTON, 8,396,800 equal-modulus paths: fidelity 95.92% (2 planes), 95.96% (3 planes), 96.77% (naive action) — Figure 1.
- Intensity correlation 97.08–97.10%; cross-check 96.06%; screen phase 5.4° ; in-mundo MAPE ≈ 0 .
- E-F3: at $T = 0$ K, exactly 1 impact position out of 400 (determinism); Pearson maximum 0.73 at 300 K, drowned to 0.24 at 4 T.
- E-F4: ghost counter-fluxes confirmed (-7.1×10^{-4} , net flux -3.4×10^{-8}).

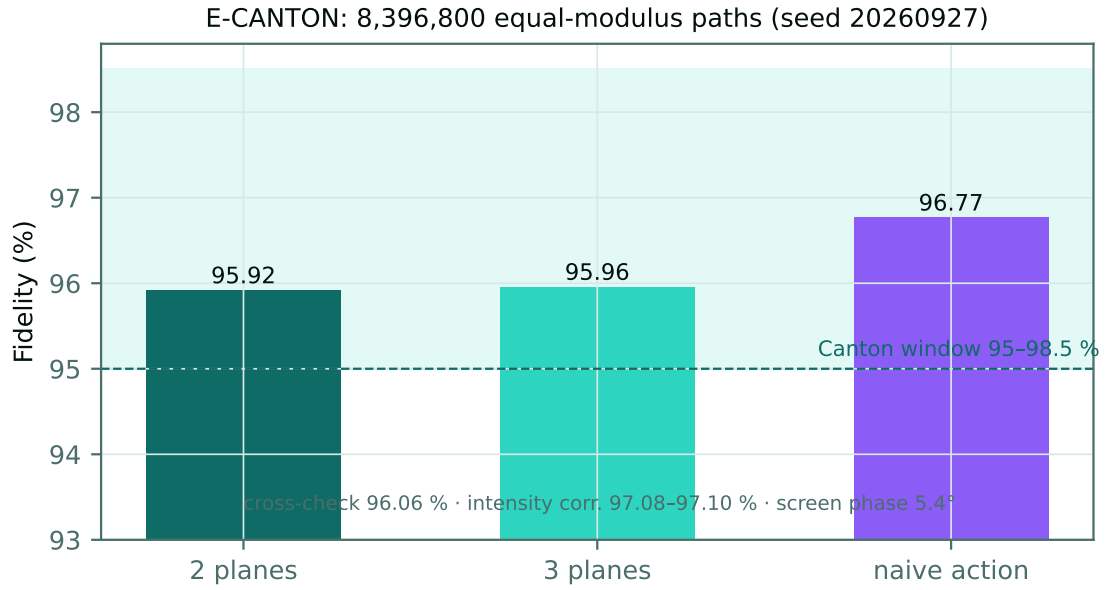


Figure 1: E-CANTON fidelity for the three action variants against the Canton window (95–98.5%). Values: 95.92 / 95.96 / 96.77%.

Negative results and falsified hypotheses

- E-F1 unresolved: sub-pixel $\pi/2$ plate (-1 px observed vs 0 predicted); a rerun at larger φ_0 is required.
- H4 half-falsified: blocking one branch produces *no* redistribution (1.00 / 0.98 / 0.96) — linearity forbids it.
- Postulate 2 not resolvable below 12° (agreement 95.9 vs 96.8%).
- E-F2: entropy 4.92 vs 4.27; vortices 326/371 (imperfect amplitude gate).

Controls and verification

Seal 33/33. Six bugs documented (B1 source at edge, B2 slits, B3 world action, B4 vortex gate, B5 2D box→paraxial engine, B6 reversed chirp 29%→95.9%). Projection mirror before export: every projected point verified by computation.

Reproducibility

Seed 20260927; campaign ≈ 1 s. Commands and datasets are in the repository (commits 2f18e80, 6f7bd1b, b455621, fb81249, de138f2, 0ccaac5).

Limitations

Computational only: no hardware validation. The paraxial world is not the physical device; agreement does not constitute a physical proof.

Data and code availability

Code and figures: <https://github.com/jonathansearch/RATISS-PHOTON> (pages: <https://jonathansearch.github.io/RATISS-PHOTON>). Primary narrative source: RATISS-ARCHIVES, POUR-LA-PROCHAINE-SESSION.md, update of 27/09 section B. Zenodo DOI: to be assigned at deposit.

References

1. Wen et al., Science Advances 12, eaeh1011 (26/08/2026).
2. RATISS-ARCHIVES, proof and recap documents, commits listed above.

RATISS Labs — independent, single-author laboratory, Yaoundé (Cameroon). No number in this document is invented: every value carries a source in Section “Data and code availability” or in the references. Negative results are published with the same prominence as positive ones.